Physics IGCSE	Summary formula	Physics grade 9
Content	**Symbols**	**System international unit (SI unit)**
1. Magnetic field due to straight line wire. **2.** Magnetic field due to Solenoid carrying electric current. **3.** Magnetic force due to d.c and a.c motor. **4.** Relationship between voltage and number of turns. **5.** Power loss via cable during transmission. **6.** Relationship between voltage and current in primary and secondary turns.	1. $B=\frac{\mu_0 I}{2\pi d}$, 2. $B=\frac{\mu_0 NI}{l}$ 3. $F=ILB$ 4. $\frac{V_S}{V_P}=\frac{N_S}{N_P}$ 5. $P_{loss}=VI=RI^2$ 6. $\frac{V_S}{V_P}=\frac{I_P}{I_S}$ $\frac{N_S}{N_P}>1$: Step.up-Transfomer $\frac{N_S}{N_P}<1$: Step.down-Transfomer	Where, B is the magnetic field measured in (T). d is the distance measured in (m). F is the magnetic force measured in (N). I is the electric current measured in (A). L is the length of wire measured in (m). N is the number of turns. V_S, V_P are the secondary and primary voltage measured in (V), respectively. I_S, I_P are the current in primary and secondary in (A), respectively. P_{loss} is the power loss measured in (W). R is the resistance of cable measured in (Ω).
7. Mirror's equation **8.** Relation between radius of curvature and focus. **9.** Magnification of mirror **10.** Object's distance **11.** Image's distance	7. $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}, f=\frac{d_o d_i}{d_o+d_i}$ 8. $f=\frac{R}{2}, R=2f$ 9. $M=\frac{d_i}{d_o}=-\frac{h_i}{h_o}, h_i=-\frac{d_i}{d_o}\times h_o$ 10. $d_o=\frac{fd_i}{d_i-f}$ 11. $d_i=\frac{fd_o}{d_o-f}$	f is the focal length measured in (m) or (cm). d_i is the image's distance measured in (m) or (cm). d_o is the object's distance measured in (m) or (cm). h_i is the height of the image measured in (m) or (cm). h_o is the height of an object measured in (m) or (cm). R is the radius of curvature measured in (m) or (cm). **Sign Convention for concave and convex mirror** • Concave mirror $(f>0)$ • Image is upright $(h_i>0)$ • Convex mirror $(f<0)$ • Image is inverted $(h_i<0)$
12. Snell's law **13.** Critical angle **14.** Relation between real and apparent depth **15.** Rarefaction index	12. $n_1 \sin i = n_2 \sin r$ 13. $\sin\theta_c=\frac{n_2}{n_1}$ 14. $n=\frac{\text{real depth}}{\text{apparent depth}}=\frac{H}{h}$ 15. $n=\frac{c}{v}, v=\frac{c}{n}$	n_1, n_2 are the rarefaction index of each medium. i is the angle of incident measured in $(^\circ)$. r is the angle of refraction measured in $(^\circ)$. θ_c is the critical angle measured in $(^\circ)$. $c=3\times10^8\ m/s$ is the speed of light. v is the speed of light in medium in (m/s).
16. Lens's equation **17.** Magnification of lens **18.** Object's distance **19.** Image's distance **20.** Sign Convention	16. $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}, f=\frac{d_o d_i}{d_o+d_i}$ 17. $M=\frac{d_i}{d_o}=-\frac{h_i}{h_o}, h_i=-\frac{d_i}{d_o}\times h_o$ 18. $d_o=\frac{fd_i}{d_i-f}$ 19. $d_i=\frac{fd_o}{d_o-f}$ 20. Converging and diverging lens	f is the focal length of lenses measured in (m) or (cm) d_i is the image's distance measured in (m) or (cm). d_o is the object's distance measured in (m) or (cm). h_i is the height of image measured in (m) or (cm). h_o is the height of an object measured in (m) or (cm). • Image is upright $(h_i>0)$ • Image is inverted $(h_i<0)$ • Converging lens $(f>0)$ • Diverging lens $(f<0)$

REVIEW FOR FINAL EXAM GRADE 9

A. Multiple choice questions

1. Which rule gives the direction of the magnetic field lines around a current carrying wire?
a. Right-hand grip rule **b.** Left-hand grip rule
c. Fleming's left-hand rule **d.** Right-hand rule

2. Which way do the arrows on the magnetic field lines around a bar magnet point?
a. To the north pole of the magnet.
b. To the north pole of the Earth.
c. To the south pole of the magnet.
d. To the south pole of the Earth.

3. Conductor carrying a current and the magnet …….
a. At right angle **b.** Magnetic field
c. Exert a force on each other **d.** Do nothing

4. The light ray that hits the mirror is called
a. Angle of incident **b.** Incident ray
c. Reflected ray **d.** Normal line

5. Which of the following is **not** a way to increase the speed at which a d.c motor turns?
a. Increasing the number of turns on the coil.
b. Increase the magnetic field.
c. Move the potential difference.
d. Decrease the current.

6. How many main types of transformers?
a. Two **b.** Three
c. Four **d.** Five

7. What type of transformer convey secondary voltage lower than primary voltage?
a. Step-up transformer **b.** Step-down transformer
c. Ideal transformer **d.** None of the above

8. What is the voltage across the secondary coil of a transformer if the primary voltage is 12 V, there are 100 turns on primary, and 300 turns on the secondary?
a. 4 V **b.** 36 V
c. 1200 V **d.** 6 V

9. A transformer has 600 primary turns connected to a 1.5 kV supply. Determine the number of secondary turns for a 240 V output voltage, assuming no losses.
a. Ns = 96 turns **b.** Ns = 69 turns
c. Ns = 79 turns **d.** None of these

10. The law of reflection states that the angle of incident
a. equals the angle of reflection **b.** Virtual
c. Rough **d.** Parallel

11. An ideal transformer with a turns ratio 2:9 is fed from a 220 V supply. Determine its output voltage.
a. V_{out} = 220 V **b.** V_{out} = 990 V
c. V_{out} = 99 V **d.** None of the above

12. A step-down transformer having a turns ratio of 20:1 has a primary voltage of 4 kV and a load of 10 kW. Neglecting losses, calculate the value of the secondary current.
a. I_s = 45 A **b.** I_s = 50 A
c. I_s = 40 A **d.** I_s = 48 A

13. How many types of beams of light travel in space?
a. Two **b.** Three
c. Four **d.** None of the above

14. How many types of main source of light?
a. Two **b.** Three
c. Four **d.** None of the above

15. What is the characteristic of image produced by pinhole camera?
a. Real image, Upright **b.** Virtual, Upright
c. Either real or virtual image, Upright **d.** Real image, Inverted

16. How many types of mirrors?
a. Two **b.** Three
c. Four **d.** None of the above

17. Which of the following is the law of reflection of light?
a. Angle of reflection is greater than angle of incidence.
b. Angle of incidence is greater than angle of reflection.
c. Angle of incidence is smaller than angle of reflection.
d. Angle of incidence is equal angle of reflection.

18. Which of the following statements about an image in a plane mirror is **incorrect**?
a. The image is laterally inverted.
b. The image is the same distance behind the mirror as the object is in front.
c. The image is real. **d.** The image is the same size as the object.

19. What is the properties of the image produced by plane mirror?
a. Real image and upright. **b.** Virtual image and erect.
c. Either real or virtual image and upright.
d. Neither real nor virtual image and inverted.

20. What is the properties of the image produced by convex mirror?
a. Image is diminished, erect and virtual.
b. Image is either large or small, upright, and virtual.
c. Image is neither large nor small, upright, and virtual.
d. None of the above

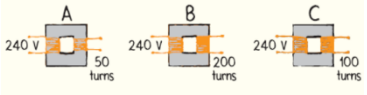
21. What is the refractive index of plastic if it slows the speed of light from 300 million m/s to 240 million m/s?
a. 1.25 million **b.** 0.8 million
c. 1.25 **d.** 0.8

22. What is the angle of refraction in a glass block of refractive index 1.52 if the angle of incidence is 32^0?
a. 32^0 **b.** 32^0
c. 45^0 **d.** 20^0

23. Light is incident in air on a glass surface at 90^0? What is the angle of refraction?
a. 90^0 **b.** 45^0
c. 0^0 **d.** 180^0

24. When light passes through a triangular glass prism, which colour is deviated through the greatest angle?
a. Red **b.** Violet
c. Green **d.** Yellow

25. The three transformers are each powered with 100 W, and all have 100 turns on the primary. The number of turns on each secondary varies as indicated.
(a) Rank the voltage output of the secondaries from greatest to least.
(b) Rank the current in the secondaries from greatest to least.
(c) Rank the power output in the secondaries from greatest to least.



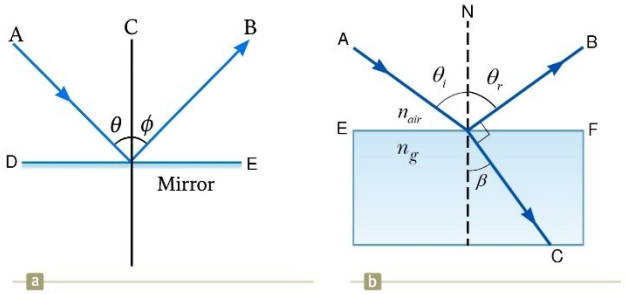
26. What is the name of the process by which light changes direction as it passes from air to glass?
a. Diffraction **b.** Refraction
c. Dispersion **d.** Reflection
27. For concave curved mirror, the incident ray that goes through the focal point will…… off the mirror parallel to the principal axis.
a. Refract **b.** Reflect
c. Incident **d.** None of the above
28. What type of image is produced when a converging lens is used as a magnifying glass?
a. Diminished **b.** Upright
c. Magnified **d.** Real
29. What type of image is produced when a converging lens is used to focus light in a camera?
a. Diminished **b.** Upright
c. Magnified **d.** Virtual
30. The right-hand grip rule for the solenoids:
a. Thumb points to direction of current, 4 fingers point to the magnetic field.
b. Thumb points to direction of force, 4 fingers point to magnetic field.
c. Thumb points to direction of magnetic field, 4 fingers point in the direction of current.
d. Thumb points to direction of force, 4 fingers point to current.
31. The space around the magnet which attracts a magnetic material is called
a. Magnet **b.** Attraction
c. Magnetic field **d.** Field lines
32. For convex lens, the incident ray that goes through the focal point will…… parallel to the principal axis.
a. Refract **b.** Reflect
c. Incident **d.** None of the above
33. The right-hand rule for straight current carrying-wire
a. Thumb points to direction of current, 4 fingers point to the magnetic field.
b. Thumb points to direction of force, 4 fingers point to magnetic field.
c. Thumb points to direction of magnetic field, 4 fingers point in the direction of current.
d. Thumb points to direction of force, 4 fingers point to current.
34. The light ray that reflect off the mirror is called
a. Angle of reflection **b.** Incident ray
c. Reflected ray **d.** Angle of incident

B. Critical thinking questions

35. State any two properties of magnetic field lines.
36. State the **Fleming Left hand rule**.
37. State **Fleming right Hand rule**.
38. What is electric motor? List down three motors.
39. What is electric generator?
40. List the factors affecting the strength of an electromagnet?
41. What is the transformer? What is the function of transformer?
42. How fast does light travel in air?
43. What is shadow? How many types of shadow?
44. How many types of mirrors and lens?
45. What is image? What is real image? Virtual image?
46. What type of image produced by plane mirror?
47. How does refractive index of a medium affect the speed of light?
48. What are lenses?

C. Work problems

49. In the diagram showing the reflection and refraction of light from a mirror, the parts are labeled with letters and angle symbols. What do points A, B, and C, the angles θ and φ, and DE Fig A represent as well as points A, B, C and N, the angle θ_i, θ_r, β and EF in Fig B represent?



50. a Copy and complete these sentences.
Electricity is distributed by the national _____. This consists of substations, cables and _____.
The electricity is transmitted at very high _____ to _____ energy being lost as heat in the cables.
The voltage can be increased or decreased by _____.
b Figure 21.20 shows a transformer. What are A, B and C representation?

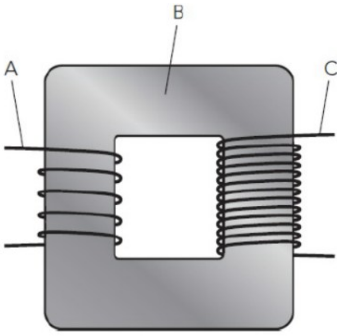


Figure 21.20: A transformer.

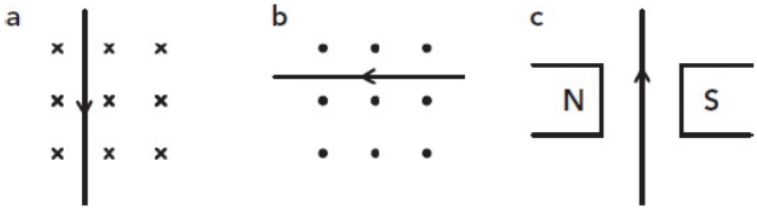
51. Copy and complete Table 21.1. Calculate the missing

N_p	N_s	V_p	V_s	Step-up or step-down transformer?
10	20		12	step-up
10		1.2	12	
	50	240	6	
10.000	20	115.000		

Table 21.1

52. A school power pack has an output voltage of 9 V. It is plugged in to the 230 V mains supply. The power pack contains a transformer. The output current of the power pack is 3 A. Calculate the current supplied to the primary coil of the transformer in the power pack. Assume there are no energy losses in the transformer.
53. A power station generates 230 000 W.
a Calculate the current in the wires when the power is transmitted:
i at 230 V **ii** at 23 000 V.
b Explain why it is better to transmit the electricity at a higher voltage.
54. There are very large transformers between power stations to the transmission cables. One of these transformers has 800 turns on its primary coil and 16 000 turns on its secondary coil. The voltage across its primary coil is 25 kV.
a State what type of transformer this is.
b Calculate the voltage across its secondary coil.

55. In Figure 20.23, crosses represent a magnetic field into the page and dots represent a field coming out of the page. In each case, use Fleming's lefthand rule to determine the direction of the force on the wire.



56. Figure 20.18 shows a type of ammeter. Copy and complete the sentences to explain how it works.

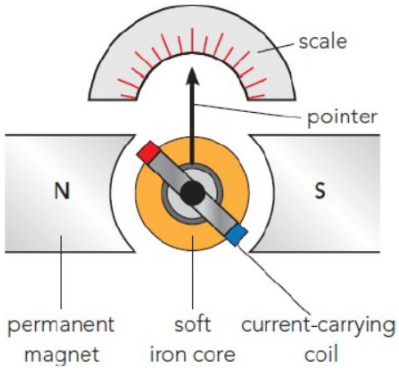
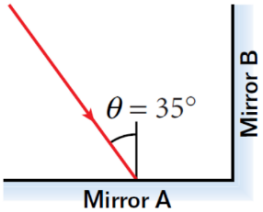


Figure 20.18:

When _____ passes through the coil, it experiences a _____ because it is in a magnetic _____. This causes the coil to rotate. When the current increases, the force will be _____ and so the pointer will move _____ on the scale.

57. Two flat mirrors make an angle of 90.0° with each other, as diagrammed at right. An incoming ray makes an angle of 35° with the normal of mirror A. Use the law of reflection to determine the angle of reflection from mirror B. What is the angle of reflection on mirror B?



58. Two plane mirrors are at an angle of $\theta_1 = 50.0°$ with each other as in the side view Shown in Figure P22.14. If a horizontal ray is incident on mirror 1, at what angle θ_2 does the outgoing reflected ray make with the surface of mirror 2?

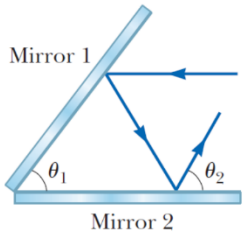
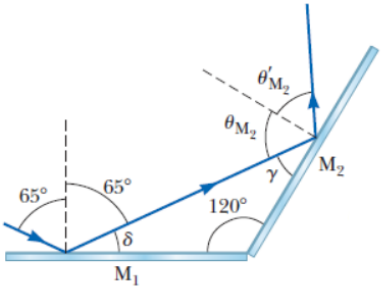


Figure P22.14

59. Two mirrors make an angle of 120° with each other as illustrated in Figure.
A ray is incident on mirror M_1 at an angle of 65° to the normal. Find the direction of the ray after it is reflected from mirror M_2.



60. Light from the Sun takes about 8.0 min to reach Earth. How far away is the Sun? Suppose the speed of light in the space between Sun and Earth is 30 000 km/s.

4. Calculate the speed of light for the following medium.

a. Water (n=1.33) **b.** Diamond (2.42)

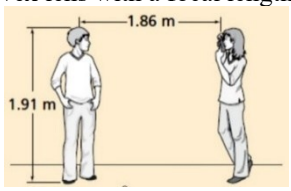
61. Calculate the refraction index for a substance if the speed of light that medium is

a. $v = 2.1\times10^8\ m/s$ **b.** $v = 1.5\times10^8\ m/s$ **c.** $v = 0.79c$

62. Calculate the index of refraction for a substance when the angle of incidence is 30^0, the angle of refraction is 50.0^0, and the index of refraction of the second substance is 1.50.

63. A beam of light passes from water into polyethylene. If i = 57.5°, what is the angle of refraction in the polyethylene? Given $n_2 = 1.50$

64. George's picture is being taken by Cami, as shown in the figure above, using a camera which has a convex lens with a focal length of 0.0470 m.
Determine George's image position.



65. A 3.0-cm-tall object is 20.0 cm from a 16.0-cm-radius concave mirror. Determine the image position and image height. Draw ray diagrams for image.

66. A concave mirror has a 7.0-cm focal length. A 2.4-cm -tall object 16.0 cm from the mirror. Determine the image height. Draw ray diagram for image.

67. An object is near a concave mirror of 10.0-cm focal length. The image is 3.0 cm tall, inverted, and 16.0 cm from the mirror. What are the object position and object height?

68. A 4.0-cm-tall object is placed 12.0 cm from a convex mirror. If the image of the object is 2.0 cm tall, and the image is located at -6.0 cm, what is the focal length of the mirror? Draw a ray diagram to answer the question. Use the mirror equation and the magnification equation to verify your answer.

69. An object is placed 30 cm from a convex lens of focal length 10 cm.

a. Calculate the image distance.

b. Is the image real or virtual?

c. Calculate the magnification of the image.

70. An object is placed to the left of a convex lens with a 25-mm focal length so that its image is the same size as the object. What are the image and object positions?

71. A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height. Draw ray diagram for image formed by lens.

72. A step-up transformer has a primary coil consisting of 200 turns and a secondary coil consisting of 3000 turns. The primary coil is supplied with an effective AC voltage of 90.0 V.

a. What is the voltage in the secondary circuit?

b. The current in the secondary circuit is 2.0 A. What is the current in the primary circuit?

73. Two separate but similar coils of wire are mounted close to each other, as shown. The first coil is connected to a battery and has a direct current flowing through it. The second coil is connected to a galvanometer.
How does the galvanometer respond when the switch in the first circuit is closed? After being closed when the current is steady? When the switch is opened?

